



[3फेज विद्युत लोकोमोटिव आई ए/आई बी/आई सी अनुसूची [आवक/जावक निरीक्षण]

**IA/IB/IC Schedule of 3Phase Electric Locomotive [Incoming/Out Going Inspection]**

लोको संख्या Loco No 32649 क्लास Class WAG9MG शिड्यूल में लेने का दिनांक Date taken under Sch 15/09/24

|                        | आगमनाIncoming / शिड्यूलSchedule |                          | अंतिमनिरीक्षणFinal Inspection |                          |
|------------------------|---------------------------------|--------------------------|-------------------------------|--------------------------|
|                        | अपर डैक<br>Upper Deck           | अंडर ट्रक<br>Under Truck | अपर डैक<br>Upper Deck         | अंडर ट्रक<br>Under Truck |
| Date/ Shift            | 15/9/24 8/H                     | 15/9/24                  | 07-10-24 8/H                  | "                        |
| Name of Technician     | Rishabh Singh                   | Nisham Khan              | Munish Chhabra                | अनुराग                   |
| Designation/ Token No. | ELF - I                         | MCF                      | MCF                           | ELF - II                 |
| Signature              | R.P. Singh                      | Rishabh - NK             | Munish Chhabra                |                          |

| Customer Feed Back/Bookings (Record feedback given by Loco pilot) |                                                                    |                             |                           |                                   |
|-------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------|---------------------------|-----------------------------------|
| Sr. No.                                                           | Customer Feed Back/Bookings                                        | Action Taken                | Name of TCN / Attended By | Remarks, if any & Sign. of JE/SSE |
| 1                                                                 | RTIS H/W                                                           | Checked ok.                 |                           |                                   |
| 2                                                                 | CCVTs mcb trip & wire fitting                                      | Same Allowed                |                           |                                   |
| 3                                                                 | Cabag ALP stat 4LT H/W                                             | attended                    |                           |                                   |
| 4                                                                 | SPM time wrong shown                                               | Checked & attend in Kanpur  |                           |                                   |
| 5                                                                 | बिगनेर site speedometer time difference                            |                             |                           |                                   |
| 6                                                                 | msg code F03/2 P2 LC-2 of bye 2 Isolate 3/4 of Transfers Available | Checked & attend by meelha. |                           |                                   |
|                                                                   |                                                                    |                             |                           |                                   |

ज.इंजी/ सी.से.इंजी.के हस्ताक्षर Signature of JE/SSE

# INSPECTION WING ELECTRICAL: CHECK AND RECORD FOLLOWINGS

| क्र.सं. कार्यवाही कार्य/निरीक्षण का विवरण / (शिड्यूल) |                                                                                                                                                 | कार्यवाही की गयी<br>Action Taken |                           | हस्ताक्षर / टिप्पणी<br>Sign/ Remarks |         |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------|--------------------------------------|---------|
| SN                                                    | Detail of Work/ Inspection (Schedule)                                                                                                           |                                  |                           |                                      |         |
| A                                                     | आगमन INCOMING (WHEN LOCO IS ENERGISED)                                                                                                          |                                  |                           |                                      |         |
| जॉब की सूची CHECK LIST                                |                                                                                                                                                 |                                  |                           |                                      |         |
|                                                       | विवरण Description                                                                                                                               | मानक मान<br>Standard Value       | वास्तविक मान Actual Value | Cab-1                                | Cab-2   |
| 1                                                     | Battery Voltage                                                                                                                                 | 100-110 V                        | 98 V                      |                                      |         |
| 2                                                     | MCB 100 Voltage                                                                                                                                 | -----                            |                           |                                      |         |
| 3                                                     | CHBA Voltage                                                                                                                                    | 95-110 V                         | 110 V                     |                                      |         |
| 4                                                     | Charging Current                                                                                                                                | 10 Amp                           | 26 A                      |                                      |         |
| 5                                                     | Working in cooling mode both Cab                                                                                                                | Working                          | cooling mode              | N/W.                                 |         |
| 6                                                     | Working of Flasher light, Marker light, Cab light, Signaling light, Indication lamp, Instrument light, Cab fan, Cab AC, Cab Heater in both cab. | Working                          |                           | working                              | working |
| 7                                                     | Working of meter UBA, U & TE/BE (Bogie-1); TE/BE (Bogie-2) by both Cab                                                                          | Working                          |                           | working                              | working |
| 8                                                     | Working of head light contactor CPR                                                                                                             | Working                          |                           | working                              | working |
| 9                                                     | Check Regression in both cab by A-9 and RS                                                                                                      | Checked                          |                           | check                                | check   |
| 10                                                    | RGCP: Opens at (cut out in kg/cm <sup>2</sup> )                                                                                                 | 10.0 ± 0.20                      | 10.0 kg/cm <sup>2</sup>   |                                      |         |
|                                                       | Close at (cut in kg/cm <sup>2</sup> )                                                                                                           | 8.0 ± 0.20                       | 7.9 kg/cm <sup>2</sup>    |                                      |         |
| 11                                                    | Working of cooling fan (Churny fan) BUR-1, BUR-2, BUR-3, BUR-4                                                                                  | Working                          | ON                        |                                      |         |
| 12                                                    | Working of cooling fan SR-1, SR-2, CEL-1, CEL-2                                                                                                 | Working                          | ON                        |                                      |         |
| 13                                                    | MRB-1 Timer ..... Second                                                                                                                        | -----                            |                           |                                      |         |
| 14                                                    | MRB-2 Timer ..... Second                                                                                                                        | -----                            |                           |                                      |         |
| 15                                                    | VCB Counting no./Pressure in kg/cm <sup>2</sup>                                                                                                 | 5.0 – 5.8                        | 5.8 kg/cm <sup>2</sup>    |                                      |         |
| 16                                                    | Condition of Silica Gel of SR & TFP Breather SR-1, SR-2, TR-1, TR-2                                                                             | Checked                          | Link                      |                                      |         |
| 17                                                    | Check Auto Flasher light by RS only and its switches.                                                                                           | Working                          |                           | check                                | check   |
| 18                                                    | Working of Flasher light by BPFL (Flasher lights are working in main & standby.)                                                                | Working                          |                           | working                              | working |
| 19                                                    | Check Working of ACP, Check Working of PVEF                                                                                                     |                                  |                           | check                                | check   |
| 20                                                    | Check and Ensure proper functioning of vigilance control device                                                                                 |                                  |                           | check                                | check   |
| 21                                                    | FDU unit set voltage at test point                                                                                                              | 0 ± 50 mv                        | -170 mv                   | Set → 35 mv                          |         |
| 22                                                    | Working of FDU by smoke to be checked the only one cab                                                                                          | Working                          |                           | working                              |         |
| 23                                                    | Working of monitor (DDS) Both cab                                                                                                               | working                          |                           | working                              | working |
| 24                                                    | Checking Brake Power (Loco should not move on 150 KN TE)                                                                                        |                                  |                           |                                      |         |

*Handwritten signature*

सं. १०१/१०१/१०१ के हस्ताक्षर Signature of JESSIE





| क्र.सं.<br>SN | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                                                                                                                                          | कार्यवाही की गयी Action Taken | हस्ताक्षर / टिप्पणी<br>Sign/<br>Remarks |            |          |     |        |     |  |     |     |       |  |  |  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------|------------|----------|-----|--------|-----|--|-----|-----|-------|--|--|--|
| <b>B</b>      | <b>GAUGES AND PRESSURE SWITCHES</b>                                                                                                                                                                                                                                               |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 1             | Ensure proper position of gauges and no difference in both cab's gauge. Check all gauge light working                                                                                                                                                                             |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 2             | Check and ensure zero error and no leakage in connection.                                                                                                                                                                                                                         |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 3             | Ensure proper tightness, working and its setting of pressure switches.                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 4             | No air leakages from inside pressure switch.                                                                                                                                                                                                                                      |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 5             | Check the electrical connectors condition and its tightness of pressure switch.                                                                                                                                                                                                   |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 6             | Check UIC socket near head light bulb.                                                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 7             | Check bulb/holder/connection on pneumatic gauge panel.                                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 8             | Check operation of corridor tube light and its switch.                                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 9             | Check fitment and operation of spot light bulb and holder in both Cab                                                                                                                                                                                                             |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 10            | Check fitment, tightness and assembly fitment of flasher unit, head light bulb connection and cable rubbing.                                                                                                                                                                      |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 11            | Visually inspect the primary current transformer resistance for evidence of overheating located on roof.                                                                                                                                                                          |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 12            | Check and clean FL LED dome and marker light.                                                                                                                                                                                                                                     |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 13            | Check all earthing points located on equipments.                                                                                                                                                                                                                                  |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 14            | Check cab light connection for tightness and cable rubbing.                                                                                                                                                                                                                       |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| <b>C</b>      | <b>Rotating Machines (MRB-1&amp;2, SCMRB 1&amp;2, SCTM 1&amp;2, OCB 1&amp;2, TMB 1&amp;2, CP-1 &amp;2, CPA), SR &amp; TFP Pump</b>                                                                                                                                                |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | <b>While Loco is energised, check the following:</b>                                                                                                                                                                                                                              |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | a) Check abnormal sound, temperature & sparking, Oil leakages and proper functioning of all auxiliary motors/Pumps.                                                                                                                                                               |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | b) Ensure that the electrical machines and its cables are adequately protected & covered thus preventing ingress of oil                                                                                                                                                           |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | c) Check direction if any motor changed.                                                                                                                                                                                                                                          |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | d) Measure the current of all four oil pumps. (IC Sch)                                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| <b>D</b>      | <b>While Loco is energised, Check and attend for proper working of</b>                                                                                                                                                                                                            |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | Working of all the rotating switches in SB-1 panel to be checked (should be in normal position)                                                                                                                                                                                   |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
|               | <table border="1"> <thead> <tr> <th>Switch no.</th> <th>Position</th> <th>Switch no.</th> <th>Position</th> </tr> </thead> <tbody> <tr> <td>154</td> <td>Normal</td> <td>160</td> <td></td> </tr> <tr> <td>152</td> <td>'0'</td> <td>237.1</td> <td></td> </tr> </tbody> </table> | Switch no.                    | Position                                | Switch no. | Position | 154 | Normal | 160 |  | 152 | '0' | 237.1 |  |  |  |
| Switch no.    | Position                                                                                                                                                                                                                                                                          | Switch no.                    | Position                                |            |          |     |        |     |  |     |     |       |  |  |  |
| 154           | Normal                                                                                                                                                                                                                                                                            | 160                           |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 152           | '0'                                                                                                                                                                                                                                                                               | 237.1                         |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| <b>E</b>      | <b>GENERAL ITEMS</b>                                                                                                                                                                                                                                                              |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 1             | Carryout detail checks on any unusual occurrences reported bydrivers in the loco log book and the same should be relayed to the shed through TLC.                                                                                                                                 |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |
| 2             | The DDS of the loco to be thoroughly examined and needful action be taken.                                                                                                                                                                                                        |                               |                                         |            |          |     |        |     |  |     |     |       |  |  |  |

*[Signature]*

हस्ताक्षर / सी.सी.ई.जी. के हस्ताक्षर Signature of JE/SSE

| क्र.सं.<br>SN | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                                 | कार्यवाही की गयी<br>Action Taken | हस्ताक्षर/ टिप्पणी<br>Sign/ Remarks |    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------|----|
| 3             | Deficiency of shunt cable of BG & Body parts.                                                                                                                            |                                  |                                     |    |
| 4             | Ensure proper functioning of auto flasher light, marker light and head light (Full/Dimmer/Rear). Check focus of Head light.                                              |                                  |                                     |    |
| 5             | Air delivery measurement in 3-phase locomotive to ascertain proper cooling and pressurization of Machine Room.<br>(As per RDSO/2009/EL/SMI/0255 Rev.'0', dtd.08.05.2009) |                                  |                                     |    |
| 6             | Check and Ensure proper connection of air dryer in SB.                                                                                                                   |                                  |                                     |    |
| 7             | Check all type of resistance and tightness of its connections.                                                                                                           |                                  |                                     |    |
| 8             | Relay 78MCR to be calibrate.                                                                                                                                             |                                  |                                     |    |
| 9             | Check Diode connection in SB Panel and check also falling of any foreign material.                                                                                       |                                  |                                     |    |
| 10            | Check tightness of Amplifier module coupler fitted between OCB-SR.                                                                                                       |                                  |                                     |    |
| 11            | Check tightness of connection and leakage of MRB capacitor.                                                                                                              |                                  |                                     |    |
| 12            | Check Earthing shunt of Bogie to Body                                                                                                                                    |                                  |                                     |    |
|               | Axle box 1-3                                                                                                                                                             | OK                               | Axle box 2-4                        | OK |
|               | Axle box 5-7                                                                                                                                                             | OK                               | Axle box 6-8                        | OK |
| 13            | Check Earthing shunt of Axle to Bogie                                                                                                                                    |                                  |                                     |    |
|               | Axle box 2                                                                                                                                                               | OK                               | Axle box 3                          | OK |
|               | Axle box 6                                                                                                                                                               | OK                               | Axle box 7                          | OK |
| 14            | Remove Electrical cable from Earth return Unit, Remove Earth return unit from Axle Box, check earth brush and replace if required.                                       |                                  |                                     |    |
| 15            | Check intactness of Axle box cover all bolts and ensure intactness of Earth return shunt from Axle box to Bogie frame.                                                   |                                  |                                     |    |
| F             | CVVRS: a) Clean dust particle deposited on all camera and NVR box.                                                                                                       |                                  |                                     |    |
|               | b) Check tightness of all connector fitted on NVR box.                                                                                                                   |                                  |                                     |    |
|               | c) Check proper function and setting of all cameras. Ensure date and time also.                                                                                          |                                  |                                     |    |
| G             | ENERGY CUM SPEED MONITORING SYSTEM (ESMON)                                                                                                                               |                                  |                                     |    |
| 1             | Visually check the EMSON for any abnormality and send the memory card to section for data retrieving.                                                                    |                                  |                                     |    |
| 2             | Energy Consumed                                                                                                                                                          |                                  | 14788899 KWH                        |    |
| 3             | Energy Regenerated                                                                                                                                                       |                                  | 1959375.5 KWH                       |    |
| 4             | Total KM                                                                                                                                                                 |                                  | 74841 KM                            |    |
| 5             | Working of Speedometer                                                                                                                                                   |                                  | working                             |    |
| H             | Simulation mode:                                                                                                                                                         |                                  |                                     |    |
| a             | Vigilance working from both cabs                                                                                                                                         |                                  | working                             |    |
| b             | 160 kept on 0 ensure speed showing 14 kmph                                                                                                                               |                                  | check ok                            |    |

  
 जे.ई.जी/ सी.सी.ई.जी.के हस्ताक्षर Signature of JE/SSE

| क्र.सं.<br>SN | कार्यवाही कार्य/निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | कार्यवाही की गयी<br>Action Taken | हस्ताक्षर/ टिप्पणी<br>Sign/ Remarks |                      |          |                     |          |                       |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------|----------------------|----------|---------------------|----------|-----------------------|----------|-------------------|--|---------------------|--|---------------------|--|--------------|--|-------------------|--|-----|----------|-----|----------|-----|----------|-----|----------|---|----------------|----|--|----|--|----|--|----|--|---|--------------|-----|--|----|--|-----|--|-----|--|---|---------------------|-----|----|-----|----|----|----|----|----|---|---------------|-----|--|-----|--|----|--|----|--|---|--------------------------------------------------------------|-----|--|-----|--|----|--|-----------------------|--|--|--|
| c             | Check both SR Contactor air leakage 12.4/1 & 12.4/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |                                     |                      |          |                     |          |                       |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| d             | FB Cubical contactor air leakage ( 8.1 & 8.2 )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | check ok                         |                                     |                      |          |                     |          |                       |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| I             | <b>Check operation of Vigilance Control Device (VCD)</b><br>VCD becomes active when any one or more operation listed below is not performed for continuous 60 seconds (1 minute)<br>i) Change of TE                      ii) Application of PVCD<br>iii) Sander operation              iv) Application of brake A-9<br>v) Application of brake SA-9<br>Check following LED Indication/Status of following items after elapse of 60 sec                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  |                                     |                      |          |                     |          |                       |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
|               | <table border="1"> <thead> <tr> <th rowspan="3">SN</th> <th rowspan="3">Indication</th> <th colspan="8">Status of Indication</th> </tr> <tr> <th colspan="2">From 60 to 68±2 sec</th> <th colspan="2">From 68±2 to 76 sec</th> <th colspan="2">After 76 sec</th> <th colspan="2">After 76 + 34 sec</th> </tr> <tr> <th>Std</th> <th>Observed</th> <th>Std</th> <th>Observed</th> <th>Std</th> <th>Observed</th> <th>Std</th> <th>Observed</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>LED Indication</td> <td>ON</td> <td></td> <td>ON</td> <td></td> <td>ON</td> <td></td> <td>ON</td> <td></td> </tr> <tr> <td>2</td> <td>Buzzer sound</td> <td>OFF</td> <td></td> <td>ON</td> <td></td> <td>OFF</td> <td></td> <td>OFF</td> <td></td> </tr> <tr> <td>3</td> <td>VCD Pneumatic Valve</td> <td>OFF</td> <td>OK</td> <td>OFF</td> <td>OK</td> <td>ON</td> <td>OK</td> <td>ON</td> <td>OK</td> </tr> <tr> <td>4</td> <td>Penalty brake</td> <td>OFF</td> <td></td> <td>OFF</td> <td></td> <td>ON</td> <td></td> <td>ON</td> <td></td> </tr> <tr> <td>5</td> <td>Re-setting of VCD by any one or more of the above operations</td> <td>YES</td> <td></td> <td>YES</td> <td></td> <td>NO</td> <td></td> <td>Only by re-set switch</td> <td></td> </tr> </tbody> </table> | SN                               | Indication                          | Status of Indication |          |                     |          |                       |          |                   |  | From 60 to 68±2 sec |  | From 68±2 to 76 sec |  | After 76 sec |  | After 76 + 34 sec |  | Std | Observed | Std | Observed | Std | Observed | Std | Observed | 1 | LED Indication | ON |  | ON |  | ON |  | ON |  | 2 | Buzzer sound | OFF |  | ON |  | OFF |  | OFF |  | 3 | VCD Pneumatic Valve | OFF | OK | OFF | OK | ON | OK | ON | OK | 4 | Penalty brake | OFF |  | OFF |  | ON |  | ON |  | 5 | Re-setting of VCD by any one or more of the above operations | YES |  | YES |  | NO |  | Only by re-set switch |  |  |  |
| SN            | Indication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                                     | Status of Indication |          |                     |          |                       |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |                                     | From 60 to 68±2 sec  |          | From 68±2 to 76 sec |          | After 76 sec          |          | After 76 + 34 sec |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Std                              | Observed                            | Std                  | Observed | Std                 | Observed | Std                   | Observed |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| 1             | LED Indication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ON                               |                                     | ON                   |          | ON                  |          | ON                    |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| 2             | Buzzer sound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | OFF                              |                                     | ON                   |          | OFF                 |          | OFF                   |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| 3             | VCD Pneumatic Valve                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OFF                              | OK                                  | OFF                  | OK       | ON                  | OK       | ON                    | OK       |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| 4             | Penalty brake                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | OFF                              |                                     | OFF                  |          | ON                  |          | ON                    |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |
| 5             | Re-setting of VCD by any one or more of the above operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | YES                              |                                     | YES                  |          | NO                  |          | Only by re-set switch |          |                   |  |                     |  |                     |  |              |  |                   |  |     |          |     |          |     |          |     |          |   |                |    |  |    |  |    |  |    |  |   |              |     |  |    |  |     |  |     |  |   |                     |     |    |     |    |    |    |    |    |   |               |     |  |     |  |    |  |    |  |   |                                                              |     |  |     |  |    |  |                       |  |  |  |

Rushabh Prasad Patil / P.P. Patil 15/9/24 08:17

Signature and Name of Technician for Inspection Incoming with date & shift:

Kamtilal Chaudhary 15/9/24 08:17  
Signature and Name of JE/SSE for Inspection Incoming with date & shift:

Maheshwar Maheshwar B. 07-10-24 08:17  
Signature and Name of Technician for Inspection Final with date & shift:

Signature and Name of JE/SSE for Inspection Final with date & shift:

अन्य किसी भी जानकारी Any Other Information:

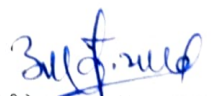
सी.से.इंजी इंचार्ज का हस्ताक्षर Signature of SSE In-charge (Inspection)

जू.इंजी/ सी.से.इंजी के हस्ताक्षर Signature of JE/SSE



# SCHEDULE PROGRESS CHART

| SN | Date    | Shift    | QC | Work Details                   | Name of JE/SSE         | Sign of JE/SSE                                                                      | Remarks |
|----|---------|----------|----|--------------------------------|------------------------|-------------------------------------------------------------------------------------|---------|
| 1  | 15/9/24 | 08<br>12 | EL | IRC to do                      | Kantilal               |  |         |
| 2  | 16/9/24 | 8-12     | EL | prox mech mecl clean sch       | Bh<br>muni<br>Kantilal |   |         |
| 3  |         |          |    | greasings to do                |                        |                                                                                     |         |
| 4  | 17/9/24 | 08<br>08 | EL | carb + 1 sch Done              | Dikson<br>Kantilal     |  |         |
| 5  | 17/9/24 | 08<br>08 | EL | FB Panel + ET sch Done         | Kantilal               |  |         |
| 6  | 19/9/24 | 08<br>08 | EL | Harmonic Filter + PT sch. Done | P. Patel               |   |         |
| 7  |         |          |    |                                |                        |                                                                                     |         |
| 8  |         |          |    |                                |                        |                                                                                     |         |
| 9  |         |          |    |                                |                        |                                                                                     |         |
| 10 |         |          |    |                                |                        |                                                                                     |         |
| 11 |         |          |    |                                |                        |                                                                                     |         |
| 12 |         |          |    |                                |                        |                                                                                     |         |
| 13 |         |          |    |                                |                        |                                                                                     |         |
| 14 |         |          |    |                                |                        |                                                                                     |         |
| 15 |         |          |    |                                |                        |                                                                                     |         |
| 16 |         |          |    |                                |                        |                                                                                     |         |
| 17 |         |          |    |                                |                        |                                                                                     |         |
| 18 |         |          |    |                                |                        |                                                                                     |         |
| 19 |         |          |    |                                |                        |                                                                                     |         |
| 20 |         |          |    |                                |                        |                                                                                     |         |
| 21 |         |          |    |                                |                        |                                                                                     |         |
| 22 |         |          |    |                                |                        |                                                                                     |         |
| 23 |         |          |    |                                |                        |                                                                                     |         |
| 24 |         |          |    |                                |                        |                                                                                     |         |
| 25 |         |          |    |                                |                        |                                                                                     |         |

  
 ज. इजी/ सी. से. इजी के हस्ताक्षर Signature of JE/SSE

## 32649 IRC Booking

- ① contactor - 218 का (mv) ज्यादा (157.5 mv) आ रहा है,
- ② हेमर वॉक्स में हेमर नहीं है।
- ③ cab#2 ALP sport light N/W. — Attend Attend
- ④ लिफ्ट जेली लिंक है।
- ⑤ machin room में दो tube light N/W. Same Allowed
- ⑥ cab#1 ALP sport light<sup>2,4</sup> N/W. — Attend
- ⑦ cab#1 tube light (LP) N/W. —

### IRC Air Flow

|                 |                      |
|-----------------|----------------------|
| TM#1 - 10.4 m/s | TM B SC#1 - 17.7 m/s |
| TM#2 - 10.9 m/s | TM B SC#2 - 17.8 m/s |
| TM#3 - 10.6 m/s | MRB#1 - 11.1 m/s     |
| TM#4 - 16.2 m/s | MRB#2 - 9.5 m/s      |
| TM#5 - 11.3 m/s | MRB SC#1 - 12.8 m/s  |
| TM#6 - 12.3 m/s | MRB SC#2 - 12.5 m/s  |

#### OCB#1

|      |      |      |
|------|------|------|
| 6.52 | 5.85 | 6.29 |
| 6.26 | 5.32 | 5.56 |
| 6.50 | 6.38 | 6.69 |

#### OCB#2

|      |      |      |
|------|------|------|
| 7.25 | 6.02 | 5.75 |
| 6.12 | 4.82 | 5.13 |
| 6.33 | 5.75 | 6.50 |

### FRC Air Flow

|             |                  |
|-------------|------------------|
| TM1 - 14.1  | TM B SC1 - 17.90 |
| TM2 - 13.50 | TM B SC2 - 18.10 |
| TM3 - 11.25 | MRB1 - 11.19     |
| TM4 - 14.20 | MRB2 - 9.80      |
| TM5 - 12.91 | MRB SC1 - 12.6   |
| TM6 - 15.60 | MRB SC2 - 9.50   |

#### OCB1

|     |      |      |
|-----|------|------|
| 7.1 | 5.80 | 6.2  |
| G1  | 6.9  | 5.58 |
| G.  | 5.98 | 5.90 |

#### OCB2

|      |      |      |
|------|------|------|
| 8    | 7.9  | 6.1  |
| 6.50 | 6.85 | 5.50 |
| 5.25 | 5.80 | 6.66 |





पश्चिम रेलवे  
Western Railway

# SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI ELECTRICAL

[3फेज विद्युत लोकोमोटिव आई ए/आई बी/आई सी अनुसूची [निरीक्षण]

IA/IB/IC Schedule of 3Phase Electric Locomotive [Inspection]

लोको संख्या Loco No 32649 क्लास Class WAB-9 शिफ्ट में लेने का दिनांक Date taken under Sch                     

| क्र.सं.<br>SN | कार्यवाही कार्य/निरीक्षण का विवरण / (शिफ्ट)<br>Detail of Work/ Inspection (Schedule)                                                                                                                                            | कार्रवाई की गयी<br>Action Taken | टीसीएन का<br>नाम<br>Name of TCN | हस्ताक्षर/ टिप्पणी<br>Sign/ Remarks |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------|-------------------------------------|
| A             | <b>डाइविंग डेस्क (Driving Desk)</b>                                                                                                                                                                                             |                                 |                                 |                                     |
| 1             | Clean the surfaces of the driver's desk with a soap solution and cloth, taking care not to allow water to enter electrical cubicles.                                                                                            | OK                              |                                 |                                     |
| 2             | Clean the dust filter on the buzzer using a soft brush without using the pressure / compressed air.                                                                                                                             | OK                              |                                 |                                     |
| 3             | Ensure that transparent rubber caps provided over the pushbuttons of driver desk.                                                                                                                                               | OK                              |                                 |                                     |
| 4             | Clean the battery, OHE and TE/BE meters in both cabs and check for zero error                                                                                                                                                   | OK                              |                                 |                                     |
| 5             | Check tightness/operation/cable rubbing of connector, earthing of angle transmitter, all rubber gasket and all equipment located on BL and its box frame inside the panel. Clean contacts, check cap and its fitment tightness. | OK                              |                                 |                                     |
| 6             | Check tightness of sub D located on BL.                                                                                                                                                                                         | OK                              |                                 |                                     |
| 7             | Check ZPT switch located on pneumatic panel.                                                                                                                                                                                    | OK                              |                                 |                                     |
| 8             | Check all earthing points located on equipment.                                                                                                                                                                                 | OK                              |                                 |                                     |
|               | <b>LIGHTING AND AIRFLOW</b>                                                                                                                                                                                                     |                                 |                                 |                                     |
| C             | <b>हेड लाइट, कैब व कोरिडोर लाइट, फ्लैशर लाइट, मार्कर लाइट HEAD LIGHT, CAB &amp; CORIDOR LIGHT, FLASHER LIGHT, MARKER LIGHT</b>                                                                                                  |                                 |                                 |                                     |
| 1             | Check lighting units for physical damage, tightness of fixing connections, particularly look for signs of overheating or burning. Replace bulbs/tubes, which have failed.                                                       | OK                              |                                 |                                     |
| 2             | Clean all internal light unit diffusers, bulbs or tubes and reflectors.                                                                                                                                                         | OK                              |                                 |                                     |
| 3             | Clean the glasses of the headlight lamp and marker light.                                                                                                                                                                       | OK                              |                                 |                                     |
| 4             | Ensure that there is no entry of water through head light lamps.                                                                                                                                                                | OK                              |                                 |                                     |
| 5             | Check for crack or breakage of the aluminum casing of headlight beams.                                                                                                                                                          | OK                              |                                 |                                     |
| 6             | Check the proper operation of flasher light, marker light (red & white), headlight, cab lights, disc light, m/c. room light.                                                                                                    | OK                              |                                 |                                     |
| 7             | Check headlight holder and clean glass & reflector. check tightness of headlight convertor. Ensure sealing                                                                                                                      | OK                              |                                 |                                     |
| 8             | Clean marker light glass and check tightness of connection. Check looseness, flashing in flasher unit.                                                                                                                          | OK                              |                                 |                                     |

Remarks if any:

*Cab & 2 Schedule attend by Shantanu Kumar*

*[Signature]*  
पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE

| क्र.सं.<br>SN | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                                                                                   | कार्यवाही की गयी<br>Action Taken | टीसीएन का<br>नाम<br>Name of<br>TCN | हस्ताक्षर/<br>दिप्पणी<br>Sign/<br>Remarks |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------|-------------------------------------------|
| <b>A</b>      | <b>Master Controller (IC Sch.)</b>                                                                                                                                                                                         |                                  |                                    |                                           |
| 1             | Calibrate the angle transmitter                                                                                                                                                                                            |                                  |                                    |                                           |
| 2             | Grease Camshafts, seats under control handles and other moving parts if stiff to operate or dry of lubricant.                                                                                                              |                                  |                                    |                                           |
| 3             | Check that the contacts are operating correctly by the camshaft and adjust if necessary.                                                                                                                                   |                                  |                                    |                                           |
| 4             | Check that the rollers are free to rotate and all wiring is sound and secure. Operate handles and ensure interlock functions correctly.                                                                                    |                                  |                                    |                                           |
| 5             | Ensure that caps have been provided to cover the auxiliary contacts.                                                                                                                                                       |                                  |                                    |                                           |
| 6             | Check the absolute value of transmitter.                                                                                                                                                                                   |                                  |                                    |                                           |
| 7             | Check the tightness of the fasteners & cable connection.                                                                                                                                                                   |                                  |                                    |                                           |
| 8             | Ensure that schematic numbers of various equipments are intact. To be provided in case of deficiency.                                                                                                                      |                                  |                                    |                                           |
| <b>B</b>      | <b>SB1 &amp; 2, HB1 &amp; 2 AND FB PANEL / CUBICLE</b>                                                                                                                                                                     |                                  |                                    |                                           |
| 1             | Check the mounting block of 8.1 and 8.2 contactors by opening side windows and check the main contactor, pre charging contactor foundation bolt tips for looseness and overheating by removing arc-chutes from front side. | OK                               |                                    |                                           |
| 2             | Check the condition of capacitors for leakage and check earth fault relay, resistance and discharging resistance for overheating. Check the tightness of all fastener lug, LEM sensor and CTs.                             | OK                               |                                    |                                           |
| 3             | Ensure that the various foundations bolts, body support bolts and supports plates of FB cubicle are intact and tight. Check tightness of cubical doors safety locks.                                                       | OK                               |                                    |                                           |
| 4             | Ensure tightness of various electrical connections, CBS in SB, HB & FB cubicles.                                                                                                                                           | OK                               |                                    |                                           |
| 5             | Check back side of HB cubicle all SB connections for tightness/overheating cable rubbing.                                                                                                                                  | OK                               |                                    |                                           |
| 6             | <b>IC Sch.:</b> a) Clean the various equipments fitted inside the SB, HB & FB cubicle.                                                                                                                                     |                                  |                                    |                                           |
|               | b) Visual examination of various EPCs & EMCs fitted inside FB cubicle and ensure that there is no crackness of cover cylinder, leakage and they are functioning properly.                                                  |                                  |                                    |                                           |
|               | c) Check the tightness of various PCB cards in central electronics.                                                                                                                                                        |                                  |                                    |                                           |
|               | d) Check the RC input filter circuit for any abnormality and measure the resistance value.                                                                                                                                 |                                  |                                    |                                           |
|               | e) Take out the arc-chutes of pre-charging contactors for looseness of strips and ensure tightness of its armature locking nut by hand.                                                                                    |                                  |                                    |                                           |
|               | f) Check the FB cable junction for tightness, cable rubbing and overheating.                                                                                                                                               |                                  |                                    |                                           |

पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE



| SN       | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                             | कार्यवाही की गयी<br>Action Taken | टीसीएन का नाम<br>Name of TCN | हस्ताक्षर / टिप्पणी<br>Sign/ Remarks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------|--------------------------------------|
| <b>C</b> | <b>HB CUBICLE :</b>                                                                                                                                                  |                                  |                              |                                      |
| 1        | Check all equipments' fitment, tightness, intactness of base foundation bolts & body support.                                                                        | OK                               | ↑                            | →                                    |
| 2        | Ensure earthing point/shunts of all panels/cubicles.                                                                                                                 | OK                               | ↑                            | →                                    |
| 3        | Check all MCB cable connections & Siemen connector tightness and no overheating.                                                                                     | OK                               | ↑                            | →                                    |
| 4        | Check tightness of 80 A & 150 A contactors connection and condition of tips.                                                                                         | OK                               | ↑                            | →                                    |
| 5        | Check 1000 V/415 V/110 V transformer connections for overheating and tightness of foundation bolts and checking of fuses for healthy connection.                     | OK                               | ↑                            | →                                    |
| 6        | Check tightness and condition of capacitor, LEM sensor below HB panel, earth fault relay & resistance connection, RLC plate, temperature sensor and all diode boxes. | OK                               | ↑                            | →                                    |
| 7        | Checking of OCB cable in HB 1 & 2 for overheating cable no. 1121, 1122, 1123 (A & B).                                                                                | OK                               | ↑                            | →                                    |
| <b>D</b> | <b>SB CUBICLE:</b>                                                                                                                                                   |                                  |                              |                                      |
| 1        | Check fasten lug connected on various equipment. Fitment, tightness.                                                                                                 | OK                               | ↑                            | →                                    |
| 2        | Check intactness of base foundation bolts & body support. Check tightness of cubical doors safety locks.                                                             | OK                               | ↑                            | →                                    |
| 3        | Check all EMC connector lug position/layout. It should not be touch with plunger.                                                                                    | OK                               | ↑                            | →                                    |
| 4        | Visually check the connections on backside of SB 1 & 2 panel for overheating of WAGO connector etc.                                                                  | OK                               | ↑                            | →                                    |
| 5        | Check tightness of DC-DC converter 24V/48V and Siemen connector for overheating.                                                                                     | OK                               | ↑                            | →                                    |
| 6        | Visually check all pressure switches, tightness of MCB, MCR relay, no voltage relay connections.                                                                     | OK                               | ↑                            | →                                    |
| 7        | Key interlocking: Check all keys are present & the complete system is operational.                                                                                   | OK                               | ↑                            | →                                    |
| 8        | Ensure the healthiness; Visually check all pre/proper functioning of rotating switch 152, 154 and 237.1 on its various locations.                                    | OK                               | ↑                            | →                                    |
| 9        | Check visually for any abnormality in SB, HB & FB cubicle                                                                                                            | OK                               | ↑                            | →                                    |
| 10       | Ensure proper functioning of thermostat of central electronics. (IC Sch.)                                                                                            | OK                               | ↑                            | →                                    |
| <b>E</b> | <b>MAIN ASSEMBLY BOX-1 &amp; BOX-2:</b>                                                                                                                              |                                  |                              |                                      |
| 1        | Clean the dust of enclosure using vacuum cleaner.                                                                                                                    |                                  |                              |                                      |
| 2        | Check the metallic strip provided for spring action in the contact is intact. Replace the contact if required.                                                       |                                  |                              |                                      |
| 3        | Check the connections for 25, 50 & 70 Sq. mm cables for their tightness. (IC Sch.)                                                                                   |                                  |                              |                                      |
| 4        | Tightness of knife contact connection (female), 3-phase choke connection, Power contactor connection. (IC Sch.)                                                      |                                  |                              |                                      |

*Signature*

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| SN       | कार्यवाही कार्य/निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (Schedule)                                                                                                      | कार्यवाही की गयी<br>Action Taken | टीसीएन का नाम<br>Name of TCN | हस्ताक्षर/<br>टिप्पणी<br>Sign/<br>Remarks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------|-------------------------------------------|
| <b>F</b> | <b>CENTRAL ELECTRONIC RACK</b>                                                                                                                                                              |                                  |                              |                                           |
| 1        | Check visually and manually all cards tightness, locking, all sub-D connection tightness, all optic fibre cable tightness for proper layout. Check tightness of cubical doors safety locks. |                                  |                              |                                           |
| 2        | Check the operation of all fans for free movements.                                                                                                                                         |                                  |                              |                                           |
| 3        | Check the tightness of connections of control cards. (IC Sch.)                                                                                                                              |                                  |                              |                                           |
| <b>G</b> | <b>FIRE DETECTION UNIT</b>                                                                                                                                                                  |                                  |                              |                                           |
| 1        | Calibrate the FDU and set voltage in between:<br>Trolex make: 0 to $\pm 100$ Mv<br><br>Siemens/Cerberus make: 0 to $\pm 50$ mV                                                              |                                  |                              |                                           |
| 2        | Check and clean FDU pipe line by vacuum cleaner. (IC Sch.)                                                                                                                                  |                                  |                              |                                           |
| 3        | Ensure the intactness / firmness of cable connection of Sicemcouplers of differential amplifiers provided below SR oil pump. (IC Sch.)                                                      |                                  |                              |                                           |

Remarks if any:

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# AUXILIARY MACHINES SCHEDULE

| क्र.सं.<br>SN | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (schedule)                                                                                                                                                       | की गयी<br>कार्यवाही<br>Action Taken | टीसीएन का<br>नाम<br>Name of<br>TCN | हस्ताक्षर/<br>टिप्पणी<br>Sign/<br>Remarks |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------------|
| <b>A</b>      | <b>Oil circulating pumps (SR &amp; TFP)</b>                                                                                                                                                                                                    |                                     |                                    |                                           |
|               | 1) Visually examine all SR&TFP oil pumps (Four nos.) for any oil leakage / any abnormal sound and take needful action.                                                                                                                         | OK                                  |                                    |                                           |
|               | 2) Check the mechanical support fasteners of all four oil pumps.                                                                                                                                                                               | OK                                  |                                    |                                           |
|               | 3) Check and secure rubbing of SR/TFP Cable.                                                                                                                                                                                                   | OK                                  |                                    |                                           |
| <b>B</b>      | <b>During schedule check all Auxiliary Blower and Motors: -</b>                                                                                                                                                                                |                                     |                                    |                                           |
|               | 1) Check tightness of foundation bolts of all auxiliary motors.                                                                                                                                                                                | OK                                  |                                    |                                           |
|               | 2) Visual examination / RDPT of impeller of OCB.                                                                                                                                                                                               | OK                                  |                                    |                                           |
|               | 3) Check the condition of all flexible hoses connected with various auxiliary blowers and replace, if required.                                                                                                                                | OK                                  |                                    |                                           |
|               | 4) Check tightness of SB and cable connections at terminal box.                                                                                                                                                                                | OK                                  |                                    |                                           |
|               | 5) Ensure that the terminal covers of various auxiliary motors are intact and water tight. Special attention needs to be given to main compressor, OCB, TMB motor cable layout and water tightness of its entry point into the motor terminal. | OK                                  |                                    |                                           |
|               | 6) Clean commutator of CPA motor. Check carbon brush and its holding assembly and replace worn out / defective carbon brushes and associated equipments.                                                                                       | OK                                  |                                    |                                           |
|               | 7) Check availability of earthing shunt.                                                                                                                                                                                                       | OK                                  |                                    |                                           |
|               | 8) Check condition of cab fan foundation, secure rubbing of wire, and check freeness of fan blade.                                                                                                                                             | OK                                  |                                    |                                           |
| <b>C</b>      | <b>IC Sch.: Besides of the above Sch. following attention needs to be given: -</b>                                                                                                                                                             |                                     |                                    |                                           |
|               | 1) Check the electrical connections of all four oil pumps.                                                                                                                                                                                     | OK                                  |                                    |                                           |
|               | 2) Check the electrical connections of all auxiliary motors.                                                                                                                                                                                   | OK                                  |                                    |                                           |
|               | 3) Greasing of all motors with recommended grease or RR-3 Servo Gem Grease .6 to 7 Stroke .15 gm.                                                                                                                                              | OK                                  |                                    |                                           |

Remarks if any:

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# ट्रैक्शन मोटर शिड्यूल TRACTION MOTOR SCHEDULE

## (A) Sch. IA/IB/IC: TM Schedule

| क्र.सं.<br>SN | कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल)<br>Detail of Work/ Inspection (schedule)                                                    | की गयी<br>कार्यवाही<br>Action Taken | टीसीएन का<br>नाम<br>Name of<br>TCN | हस्ताक्षर/<br>टिप्पणी<br>Sign/<br>Remarks |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------------|
| 1             | Examine all traction motors for signs of damage, dents or other defects caused by ballast. Check air outlets are not obstructed in any way. | OK                                  |                                    |                                           |
| 2             | Check traction power cables, speed sensor and temperature sensor cables and ensure that they are not chamfered or damaged in any way.       | OK                                  |                                    |                                           |
| 3             | Check Tightness of both end cover bolts of Traction Motor.                                                                                  | OK                                  |                                    |                                           |
| 4             | Check the intactness of the junction box cover and bolt                                                                                     | OK                                  |                                    |                                           |
| 5             | Open TM Junction box at body side and check the tightness of connections                                                                    | OK                                  |                                    |                                           |
| 6             | Check the condition of bellows and replace if required.                                                                                     | OK                                  |                                    |                                           |

## (B) Sch. IC: TM Schedule & Greasing

| क्र.सं.<br>SN | कार्यवाही कार्य / का विवरण निरीक्षण / (शिड्यूल)<br>Detail of Work/ Inspection (schedule)                                                                                                      | की गयी<br>कार्यवाही<br>Action Taken | टीसीएन का<br>नाम<br>Name of<br>TCN | हस्ताक्षर/<br>टिप्पणी<br>Sign/<br>Remarks |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------------|
| 1             | Check for intactness of Grease nipple on NDE & DE side. Lubricate the both end bearings using a grease gun with specified grease and send old grease sample to Lab for testing metal content. |                                     |                                    |                                           |
| 2             | Check the condition of bearing by seeing the grease, which comes out from grease inlet. Check the condition of grease too.                                                                    |                                     |                                    |                                           |
| 3             | Send grease sample to Lab for testing metal content, whenever grease available.                                                                                                               |                                     |                                    |                                           |
| 4             | Check under truck TM cables for any rubbing against metallic body and secure cables properly.                                                                                                 |                                     |                                    |                                           |
| 5             | If applicable following to be check:                                                                                                                                                          |                                     |                                    |                                           |
|               | i) Ensure tightness of all bolts of all Hurth coupling.                                                                                                                                       |                                     |                                    |                                           |
|               | ii) Check oil level of Hutch coupling and send sample of oil to Laboratory.                                                                                                                   |                                     |                                    |                                           |
|               | iii) Check all TM coupling membrane for crack and oil leakages.                                                                                                                               |                                     |                                    |                                           |
|               | iv) Clean sleeve and star of Hutch coupling.                                                                                                                                                  |                                     |                                    |                                           |
|               | v) Check and replace bolt if found defective treads of bolts of sleeve and star of Hutch coupling.                                                                                            |                                     |                                    |                                           |
|               | vi) Check free movement of Hutch coupling.                                                                                                                                                    |                                     |                                    |                                           |



| Sl. No. | Details of Work Inspection (Schedule)                                                                                                                                | Action Taken | Name of I.C.N | Sign / Remarks |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|----------------|
| 1       | <b>HARMONIC FILTER</b>                                                                                                                                               |              |               |                |
| 2       | Examine the roof mounted resistor for any signs of damage or overheating or presence of foreign materials and clean debris away from the resistor and its insulation | OK           | Rip prigo Red | P.P. Red       |
| 3       | Check and ensure the electrical connections and its tightness                                                                                                        | OK           |               |                |
| 4       | Visually inspect the contact tips of contactors for signs of any damage. Repair / replace the defective parts                                                        | OK           |               |                |
| 5       | Visually inspect the capacitor elements inside the cubicle checking for signs of damage                                                                              | OK           |               |                |
| 6       | Check and Record capacitor Bank capacitance                                                                                                                          | OK           |               |                |
| 7       | Check insulation (IR) value - 300                                                                                                                                    | OK           |               |                |
| 8       | (X) (X) Blow & clean the filter resistor and its insulation                                                                                                          | OK           |               |                |
| 9       | (X) (X) Perform functional checks ensuring the main and Aux circuit to operate correctly                                                                             | -            |               |                |
| 10      | <b>PROTECTION SYSTEM - TRACTION CONVERTER</b>                                                                                                                        |              |               |                |
| 1       | Check the cable connection of Niche connectors of differential amplifier provided below SN oil pump                                                                  | OK           | Chaitanya     | Chaitanya      |
| 2       | Check the condition of silico gel. If pink replace with dried silicagel                                                                                              | OK           |               |                |
| 3       | (X) Clean capacitor circuit capacitor.                                                                                                                               | -            |               |                |
| 4       | (X) Cleaning by Vacuum cleaner                                                                                                                                       | -            |               |                |
| 5       | (X) Clean the bank capacitors by blowing after opening side cover.                                                                                                   | -            |               |                |
| 6       | Check that there is no sign of leakage from the capacitors                                                                                                           | -            |               |                |
| 7       | (X) Clean the isolating blade and spring contact of earthing switch.                                                                                                 | -            |               |                |
| 8       | Lightly lubricate with the specified grease                                                                                                                          | -            |               |                |
| 9       | Check the tightness of SN electronic P.H                                                                                                                             | -            |               |                |
| 10      | Ensure sealing of incoming cable gland of control card rack to avoid dust entry                                                                                      | -            |               |                |
| 1       | Check the condition of parts of traction converters                                                                                                                  | -            | Chaitanya     | Chaitanya      |
| 2       | <b>ACCESSORY CONVERTER 1, 2 &amp; 3</b>                                                                                                                              |              |               |                |
| 1       | Visually check the condition of cable and terminations for overheating                                                                                               | -            |               |                |
| 2       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 3       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 4       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 5       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 6       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 7       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 8       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |
| 9       | Check the condition of cable and terminations for overheating                                                                                                        | -            |               |                |

Chaitanya

| F TRACTION CONVERTER (SR) |                                                                                                                                                                                                                                                                                          |    |              |              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------|--------------|
| 1                         | Examine all electrical equipment of traction converter for signs of dirt, corrosion, damage etc. Remove all dust/dirt deposits from the connection insulators. Check the tightness of foundation supports and side supports for crackness. Check tightness of cubical doors safety locks | OK |              |              |
| 2                         | Check the main contactor, pre-charging contactor foundation bolt and tips for looseness and overheating by removing archutes. Check and ensure tightness of pre-charging contactor armature locking nut and strips                                                                       | OK |              |              |
| 3                         | Check all connected fast on lug, bus bar connection, CTs, all LEM sensor connections, gate unit connections, pre-charging, MUB and its earth fault resistance, DC link and series resonance capacitor for leakage, tightness and overheating                                             | OK |              |              |
| 4                         | Check the connection of earth fault relay. DC link sensor and all the earthing point of SR1, SR2                                                                                                                                                                                         | OK |              |              |
| 5                         | Check BDV of traction converter oil (in GTO based loco), should be 30KV                                                                                                                                                                                                                  | —  |              |              |
| 6                         | Carry out DGA of traction converter oil by C&M Lab (in GTO base loco)                                                                                                                                                                                                                    | —  |              |              |
| 7                         | Check the oil level indicator situated on the conservator(Expansion Tank). If the oil is below the minimum mark-top up with the specified oil. Check for any signs of leakage, Range Oil level middle strip +/- 1.5"                                                                     | OK | RAJESH K. K. | RAJESH K. K. |
| 8                         | Check for any leakage, loose, missing of screw from oil pipe joint, two flange joints, wire braided pipe, valve set vent plug, stoochi coupling, top bottom oil pipe, back body of valve set                                                                                             | OK |              |              |
| 9                         | SR-1 & SR-2 terminal, if difference of inductance between any two phase is < 0.005mH, allow the motor for next schedule                                                                                                                                                                  | —  |              |              |
| 10                        | If the value is more than 0.005 mH, measure the inductance of individual TM and if difference between two phase of motor is < 0.015mH, allow the motor for next schedule. If > 0.015mH, remove the motor                                                                                 | —  |              |              |
| 11                        | Check the cable connection of Sichem connectors of differential amplifier provided below SR Oil pump.                                                                                                                                                                                    | OK |              |              |
| 12                        | Check ventilation fan of Traction converter.                                                                                                                                                                                                                                             | OK |              |              |

  
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